

PROBLEM

A computational arbitration biomarker for hallucination risk in Parkinson's disease

August, 2026

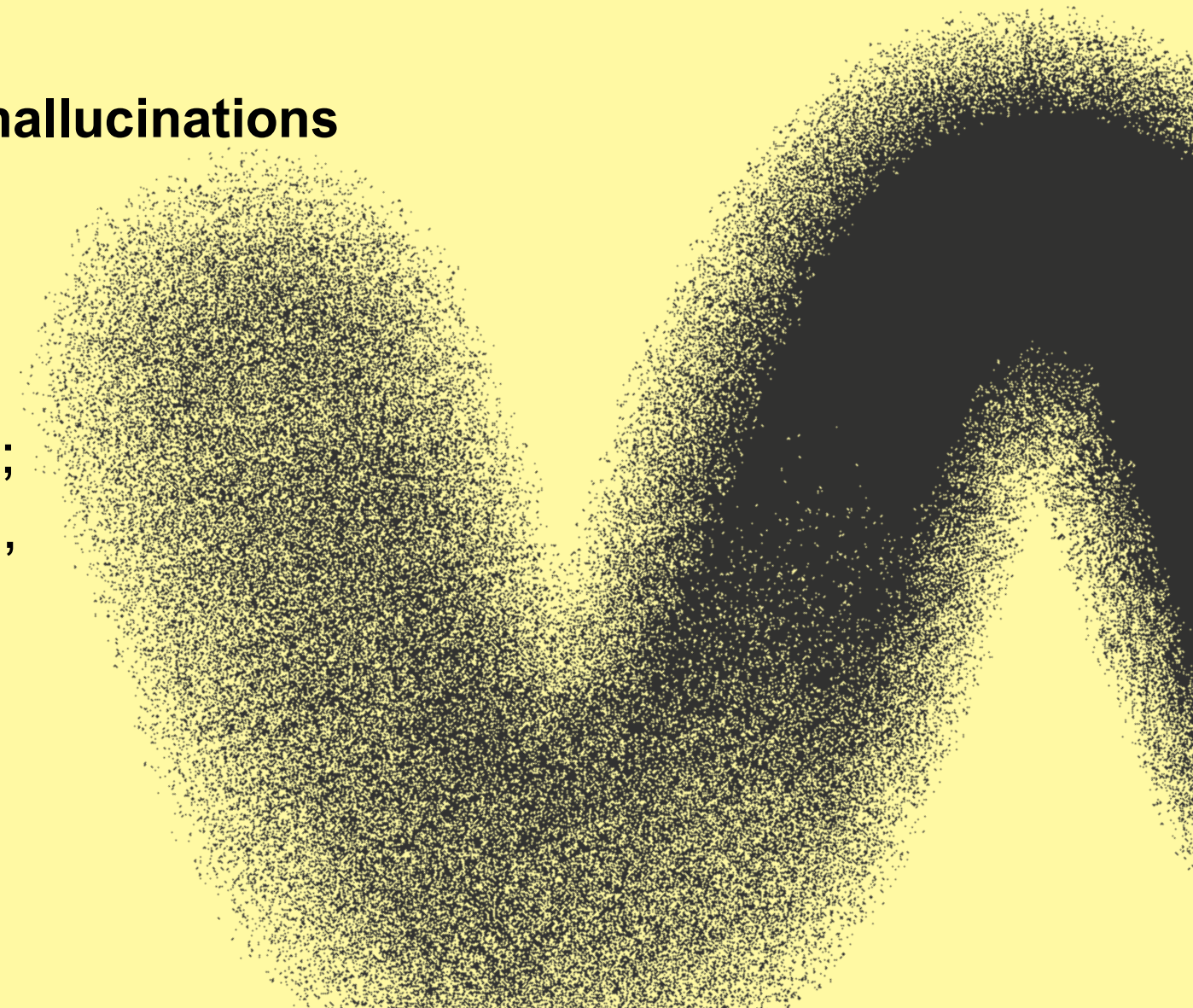
Dung Tran | Rayo Lab Pte. Ltd.

~60%

of people with Parkinson's disease develop hallucinations and no clinical test predicts who, or when.

Predictive-coding theory: hallucinations can show a wrong match between brain guesses and real senses (Rao & Ballard, '99; Friston, '10; Zarkali et al., '19; Adams et al., '13, Sterzer et al., '18).

We asked if that idea could work as an early warning signal in large clinical dataset.



Teaching a model to catch you guessing

HALLUCINATION	D'	TOTAL FA	RELATED FA	UNRELATED FA	FA RATIO	N
None	3.95	1.00	0.84	0.16	5.25	4,038
Present	3.42	1.56	1.17	0.39	2.97	94
Frequent	3.09	2.02	1.34	0.68	1.96	41

Switching-observer model reveals a breakdown in arbitration where sensory sensitivity falls and guessing from prior expectations increases. This insight is extracted directly from routine clinic **HVLT-R recognition data** using signal detection theory without needing new instrumentation.

↓ 22% d', None → Frequent 4.25× unrelated FA, None → Frequent

PD patients only, grouped by hallucination severity

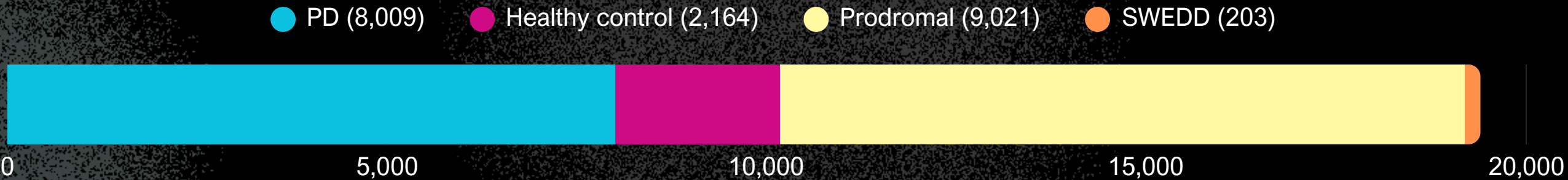
5,066

SUBJECTS

19,397

CLINICAL VISITS

PPMI cohort, deduplicated visit counts



$d' = \Phi^{-1}(\text{hit rate}) - \Phi^{-1}(\text{false-alarm rate})$

Standard signal-detection sensitivity

how well responses track what was actually shown.

$c = -\frac{1}{2} [\Phi^{-1}(\text{hit rate}) + \Phi^{-1}(\text{false-alarm rate})]$

Response criterion

a general "yes" bias, independent of sensitivity.

$P(\text{yes}) = p \cdot P_{\text{sensory}}(\text{yes}) + (1 - p) \cdot P_{\text{prior}}(\text{yes})$

The switching mixture

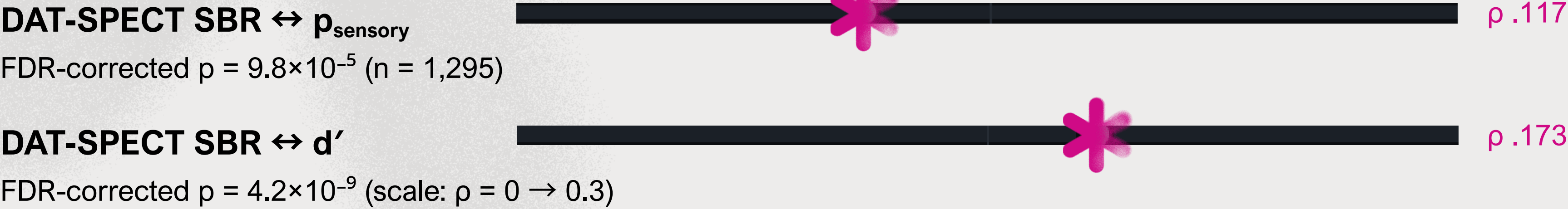
real answers blend a d'/c-driven process with a fixed plausibility habit (p is the arbitration parameter we report throughout).

During the test, the computer/ interviewer shows the subject 36 words, one at a time . The subject says "yes" if they remember the word or "no" if they do not.

Each person ends up with three counts after the test: how many "yes"es out of the 12 targets (that's their hits), how many "yes"es out of the 12 related words (related false alarms), how many "yes"es out of the 12 unrelated words (unrelated false alarms).The model only sees these three raw data counts (0 to 12) per person (HVLT-R).

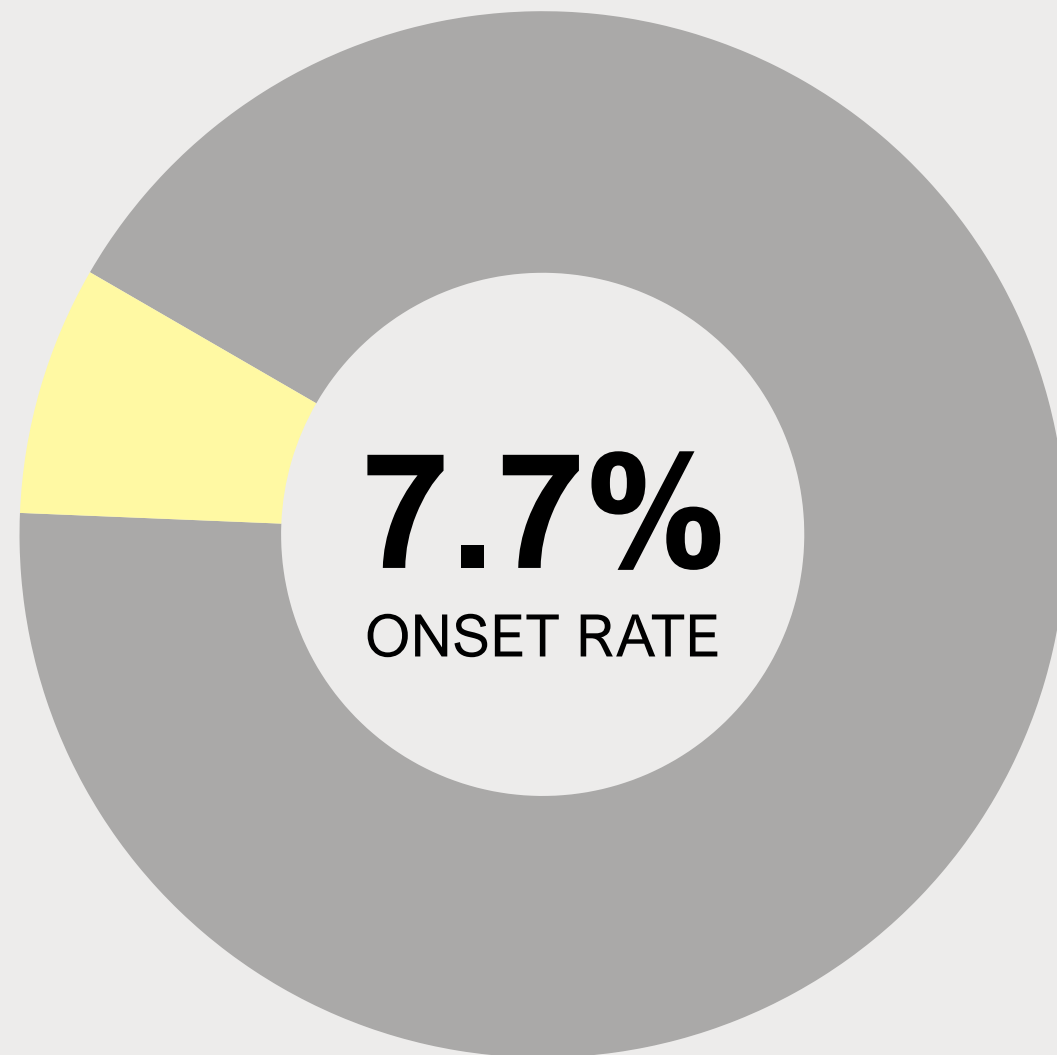
Type (12 each)	Definition	Example	Metric ("Yes" Count)
Target	On the studied list	Studied "DOCTOR", tested "DOCTOR"	Hits (0–12)
Related Distractor	Not on list, same theme	Studied "DOCTOR", tested "NURSE"	Related False Alarms (0–12)
Unrelated Distractor	Not on list, no connection	Studied "DOCTOR", tested "BICYCLE"	Unrelated False Alarms (0–12)

It tracks dopamine loss and *replicates*



Replicates held out 50/50 split of patients: $p = .141$ ($p < .001$) and $p = .086$ ($p = .03$) -> same sign, both significant, independently.

It predicts *who* before hallucinations appear



Survival analysis, unaffected by the merge-duplication correction

77 of 996 hallucination-free PD patients developed hallucinations during follow-up, measured before clinical reporting,.

OR **.795** per unit d' ($p = .025$)

Lower baseline sensory sensitivity
→ higher future risk.

*Single-visit parameter recovery is noisy ($r = 0.226$).
We are pooling multi-visit information to sharpen it
and reporting these preliminary results now.*

Modest, real, and built to scale



Replicates

Held-out 50/50 split
significant in both halves,
same direction.



Predicts

Flags risk before
hallucinations are ever
clinically reported.



Scalable

Built from a memory test
clinics already run
without new hardware.

Next sharpen precision across visits, then explore remote and
wearable monitoring built on the same signal.

Rao, R. P. N., & Ballard, D. H. (1999). *Predictive coding in the visual cortex: A functional interpretation of some extra-classical receptive-field effects.* **Nature Neuroscience**, 2(1), 79–87

Friston, K. (2010). *The free-energy principle: a unified brain theory?* **Nature Reviews Neuroscience**, 11(2), 127–138.

Zarkali, A., et al. (2019). *Increased weighting on prior knowledge in Lewy body-associated visual hallucinations.* **Brain Communications**, 1(1), fcz007.

Adams, R. A., et al. (2013). *The computational anatomy of psychosis.* **Frontiers in Psychiatry**, 4, 47.

Sterzer, P., et al. (2018). *The predictive coding account of psychosis.* **Biological Psychiatry**, 84(9), 634–643.

Green, D. M., & Swets, J. A. (1966). *Signal detection theory and psychophysics.* **John Wiley.**

THANK YOU!

Laquitaine, S. & Gardner, J.L. (2018). *A switching observer for human perceptual estimation.* **Neuron**, 97(2), 462–474.

Parkinson's Precision Medicine Initiative. (2026). Parkinson's Precision Medicine Initiative (PPMI) *Dataset.* **Laboratory of Neuro Imaging (LONI) Image and Data Archive.** <https://ida.loni.usc.edu/>